bstok.ms 26

AI

ChatGPT

AI

ChatGPT

AI

Claude

ChatGPT

AI

Gemini

Claude

ChatGPT

Cohere

Gemini

AI

FLUX

Black Forest

Meta

LLaMA

Claude

Qwen

Qwen

Phi

Microsoft

XAI

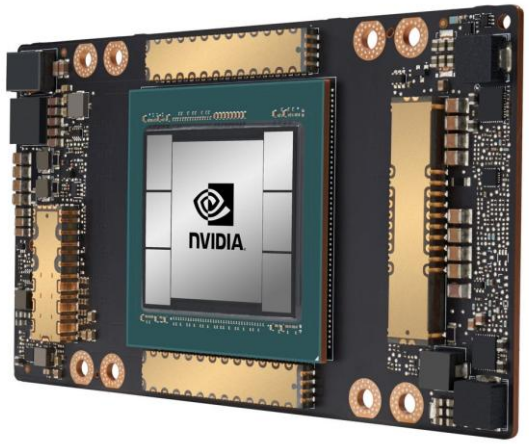
DeepSeek

Mistral

ChatGPT

Gemini

AI



Claude

ChatGPT

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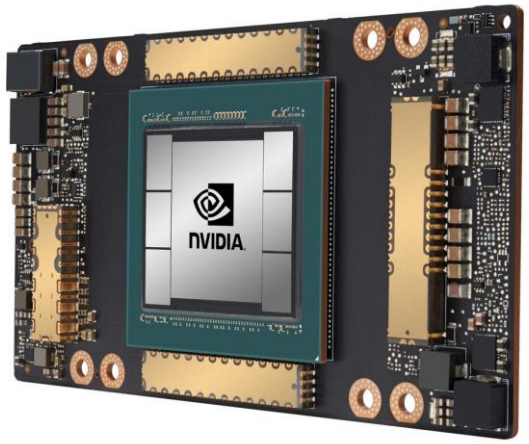

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Gemini

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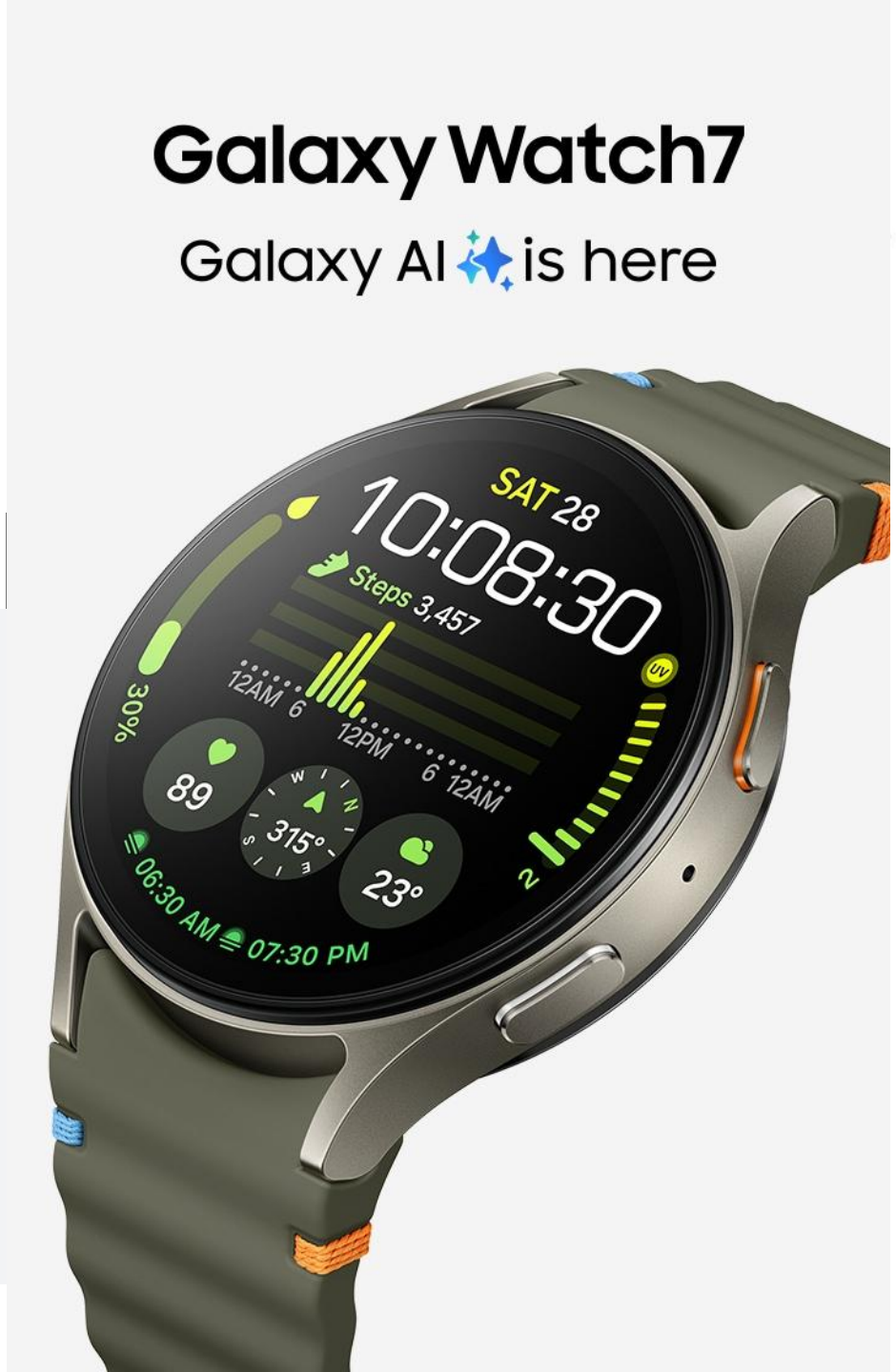
Samsung Vision AI

A


Galaxy Watch7
Galaxy AI is here
SAT 28
10:08:30
Steps 3,457
12AM 6 12PM 6 12AM
30%
89
315°
23°
06:30 AM
07:30 PM

Samsung Vision AI

Family Hub™ AI

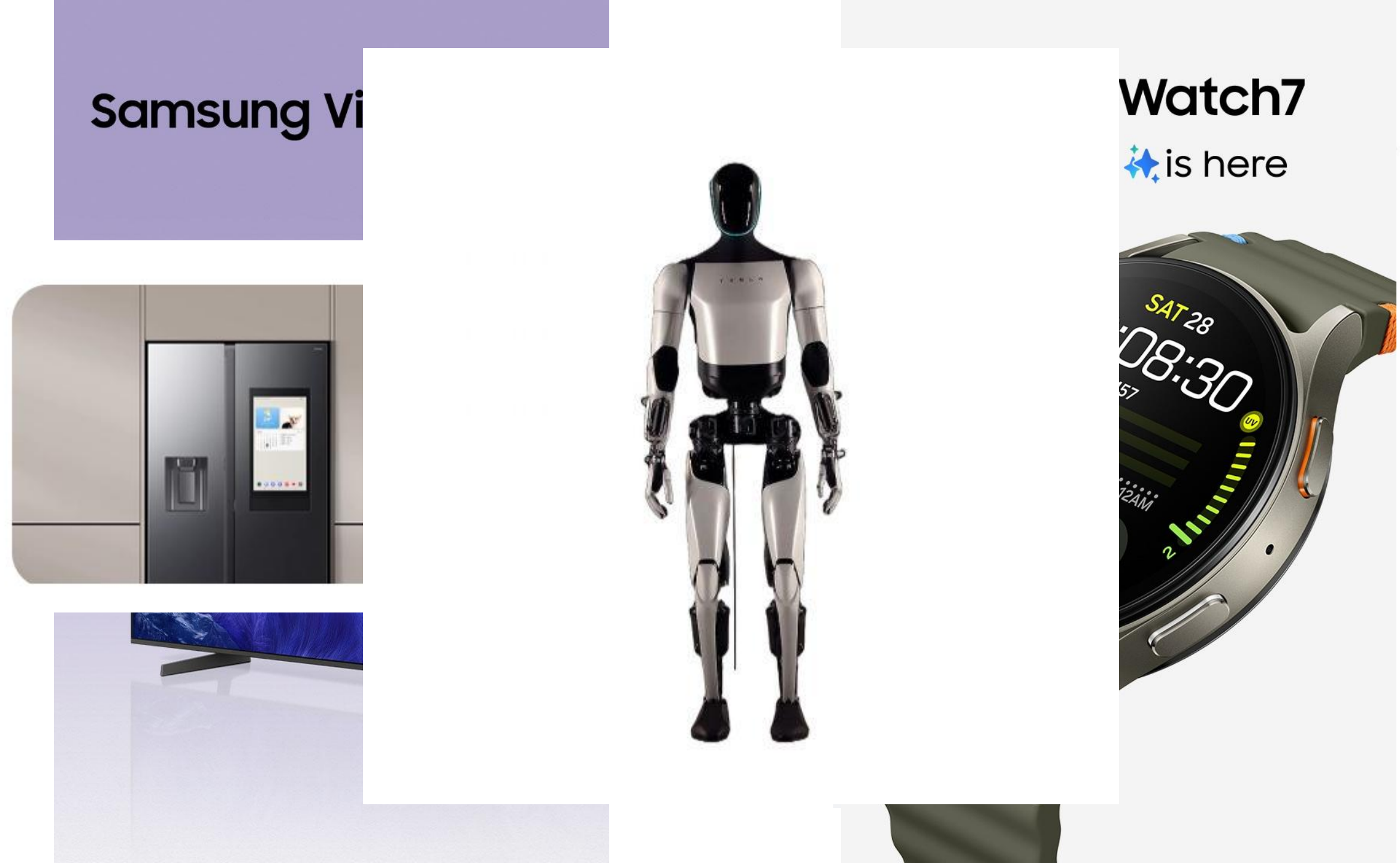
Lodówka ze zintegrowanym ekranem, łącznością Wi-Fi oraz dedykowanymi aplikacjami.

View Inside

Na ekranie smartfona możesz sprawdzić aktualną zawartość lodówki.

Galaxy Watch7

Galaxy AI is here


Samsung Vi
Watch7
is here
SAT 28
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12AM

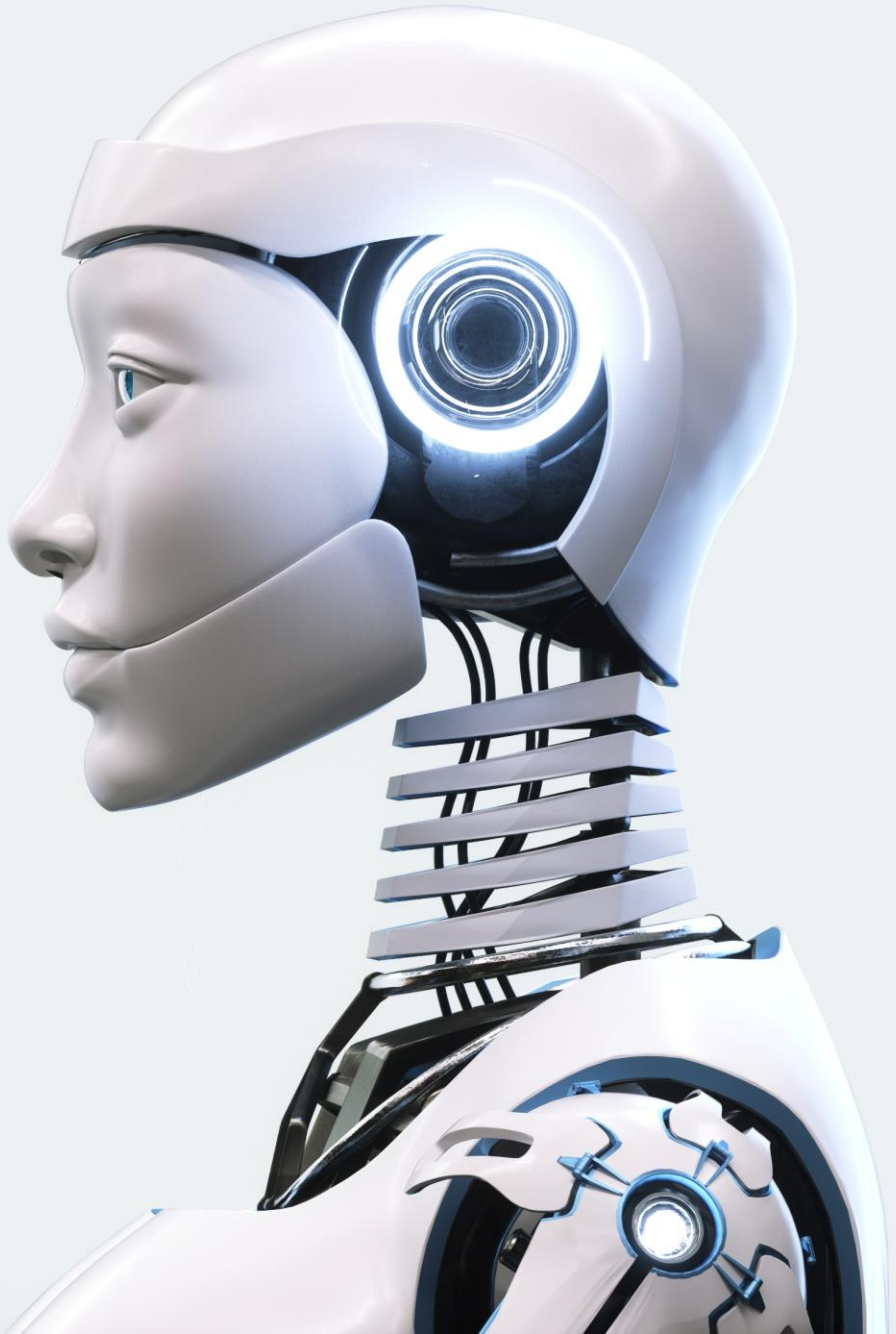
IDOL

IDOL

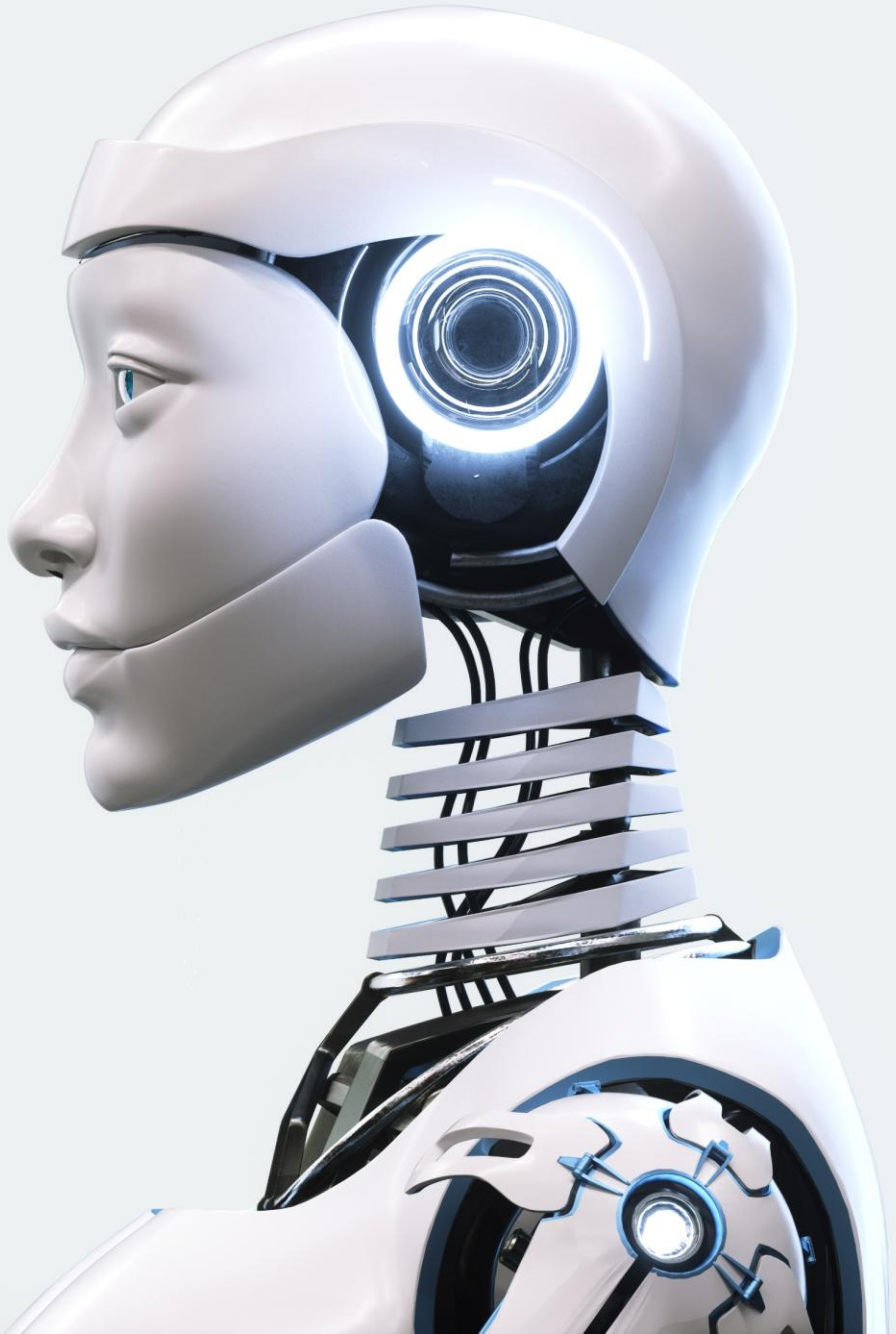
IDOL

AI

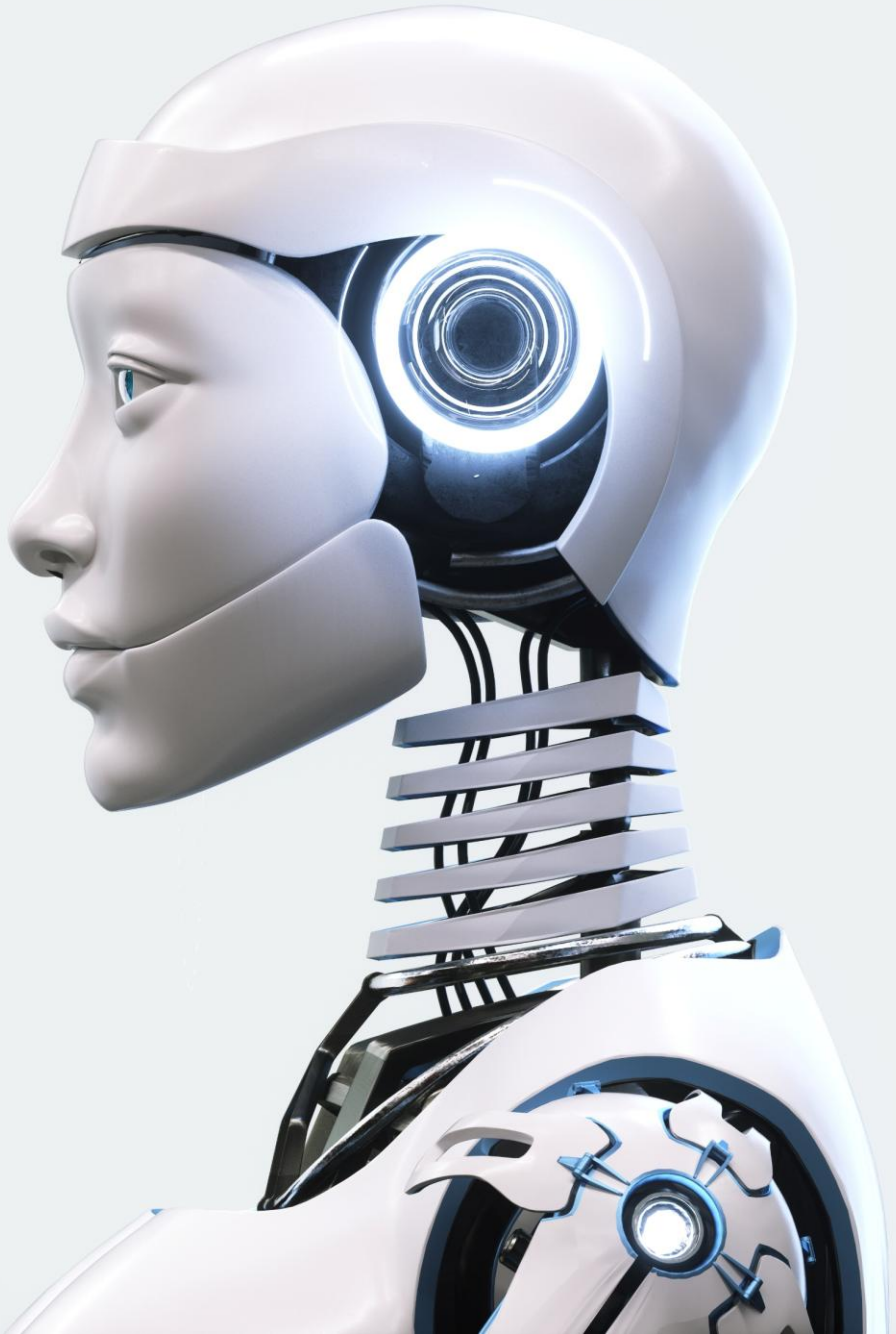
AI WILL STILL OUR WORK



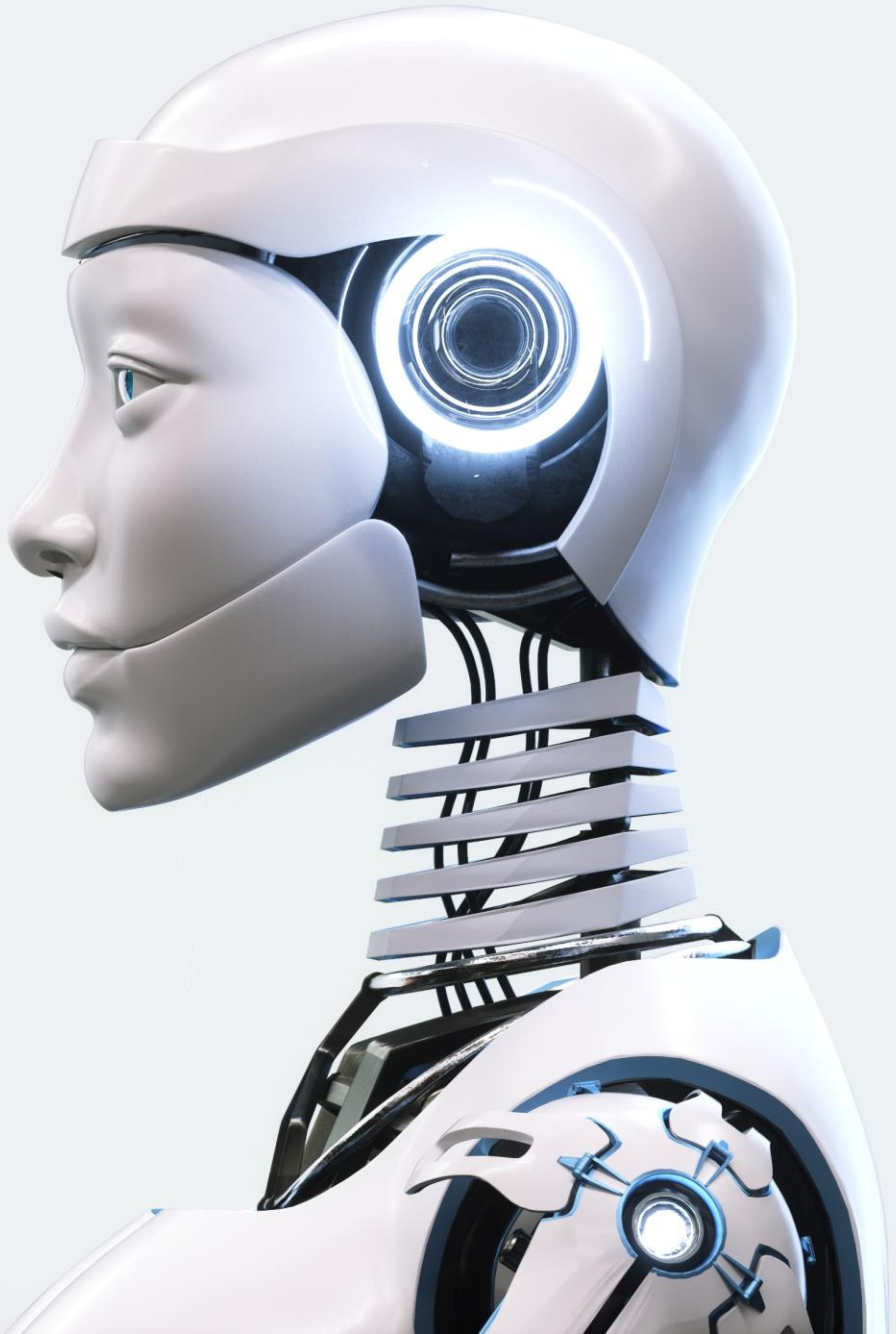
AI WILL CHANGE OUR WORK



HOW AI CHANGE OUR WORK



HOW AI CHANGED MY WORK



Designing Software Architecture

for AI Agent

Designing Software Architecture for AI Agent

Sebastian Dąbkowski

Sebastian Dąbkowski

.NET 16 lat

Sebastian Dąbkowski

.NET 16 lat

Architecture 8 lat

Sebastian Dąbkowski

.NET	16 lat
Architecture	8 lat
Coding AI	9

Sebastian Dąbkowski

.NET	16 lat
Architecture	8 lat
Coding AI	9 miesięcy

Sebastian Dąbkowski

.NET	16 lat
Architecture	8 lat
Coding AI	9 miesięcy

CEO &
Founder

ARPEGGIA
.STUDIO

Sebastian Dąbkowski

Designing Software Architecture for AI Agent

Sebastian Dąbkowski

Let me tell you...

Let me tell you...

a story

Let me tell you...

a story

about an imaginary client

Let me tell you...

a story

about an imaginary client

based on a proven process

MAIL

Temat: Zapytanie o realizację systemu „Mercato” – marketplace dla sklepów internetowych
Od: Jan Kowalski jan.kowalski@example.com
Do: Sebastian Dąbkowski

Cześć Sebastian,
chciałbym zlecić stworzenie systemu o nazwie **Mercato**. Ma to być **platforma typu marketplace**, która połączy różne sklepy internetowe w jednym miejscu.
Zależy mi, żeby Mercato miało dwa główne obszary:

1. **Panel sprzedawcy** – miejsce, gdzie sklepy mogą się rejestrować, dodawać produkty, zarządzać zamówieniami, płatnościami, wysyłkami i obsługą klientów. Każdy sklep powinien mieć własny mini-panel administracyjny, ale cały system działa w ramach jednej wspólnej platformy.
2. **Strona dla kupujących** – portal, na którym użytkownicy mogą przeglądać oferty różnych sklepów, dodawać produkty do koszyka, kupować je, wystawiać opinie i śledzić swoje zamówienia.

Funkcjonalnie chciałbym, żeby Mercato oferowało pełny zestaw funkcji, jakie ma współczesny marketplace:

- rejestracja i logowanie użytkowników (kupujący, sprzedawcy, administratorzy),
- wystawianie i edycja produktów (zdjęcia, opisy, ceny, stany magazynowe),
- wyszukiwanie i filtrowanie ofert,
- koszyk zakupowy, płatności online, historia zamówień,
- obsługa wysyłek i integracja z firmami kurierskimi,
- system ocen i komentarzy,
- panel administracyjny do moderowania ofert i użytkowników,
- powiadomienia e-mail i webowe,
- statystyki sprzedaży i raporty dla sprzedawców.

Zależy mi, żeby system był przygotowany tak, by można go było łatwo rozwijać o kolejne moduły (np. promocje, programy lojalnościowe, integracje z zewnętrznymi sklepami przez API).
Chciałbym, abyś pomógł mi zaplanować architekturę rozwiązania, określił etapy realizacji i przedstawił wstępny harmonogram prac.

Pozdrawiam,
Jan Kowalski
Założyciel projektu Mercato

MEETING

Business Scope

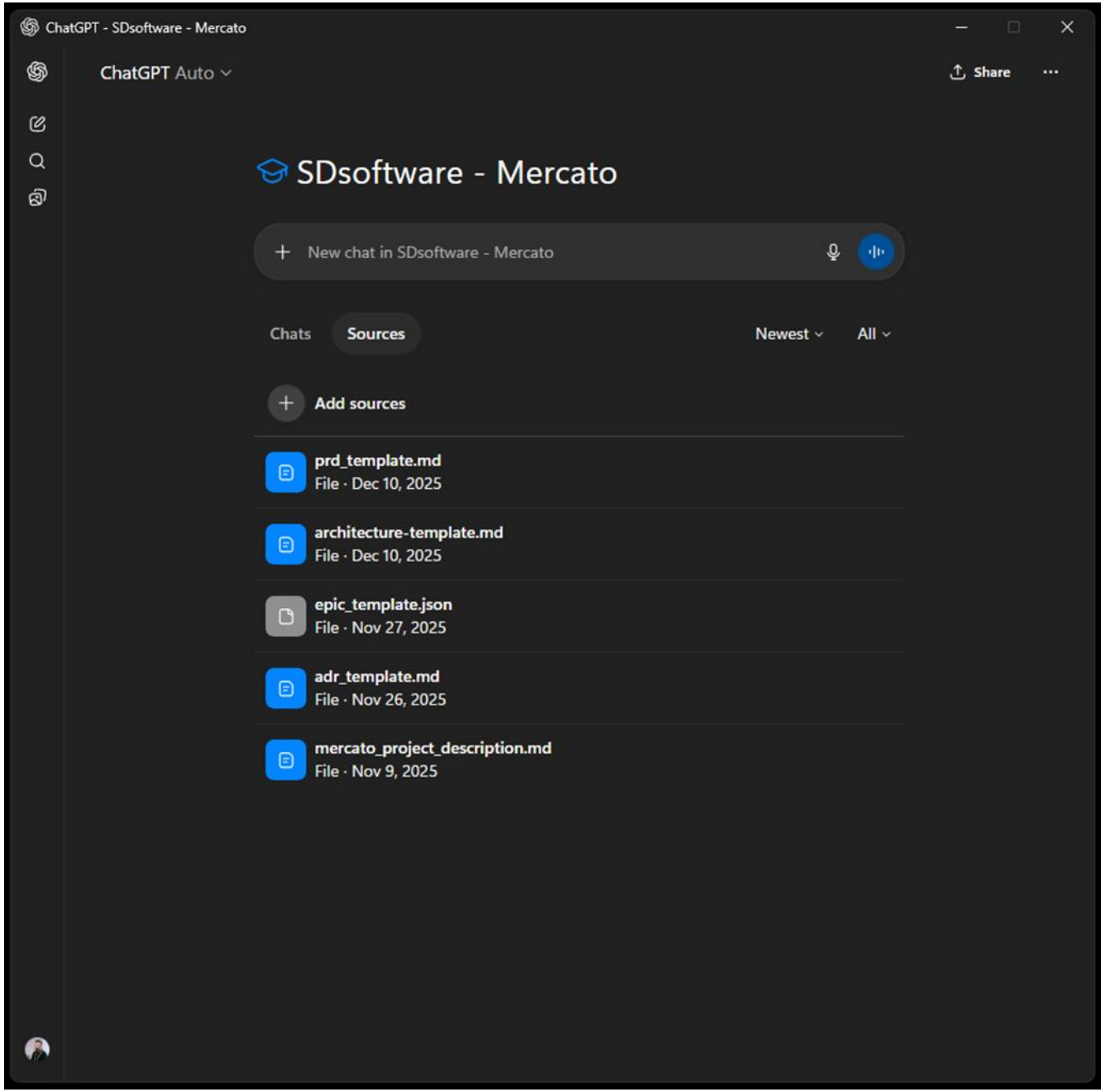
Features | Revenue | Costs | How? | What? | Who?

ChatGPT

Instruction

Templates

Project Info



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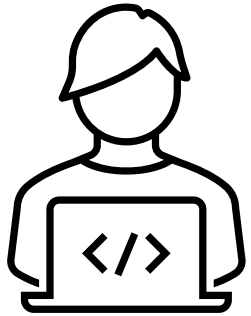

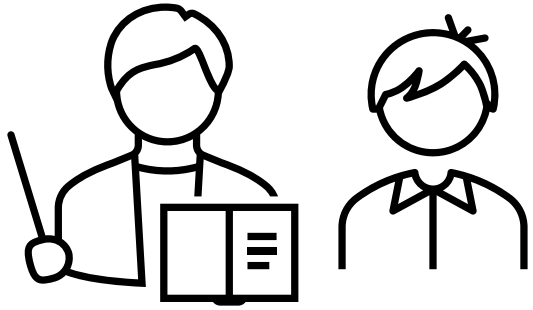

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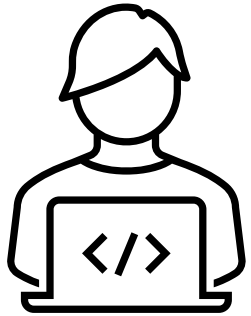

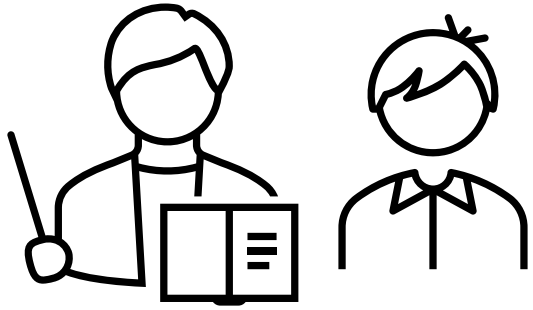

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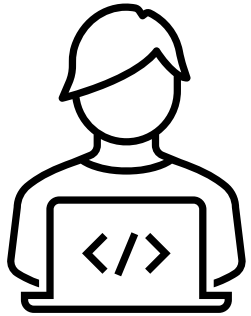
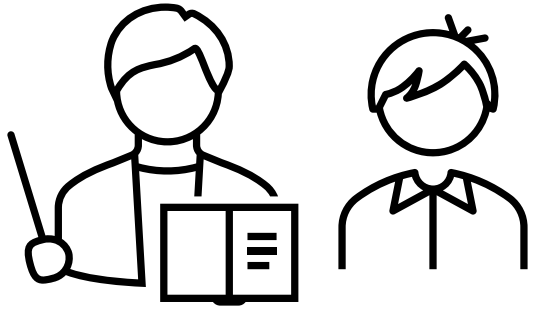
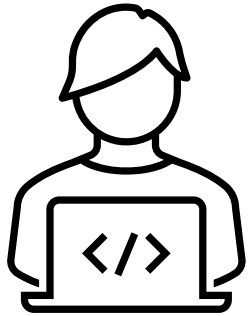


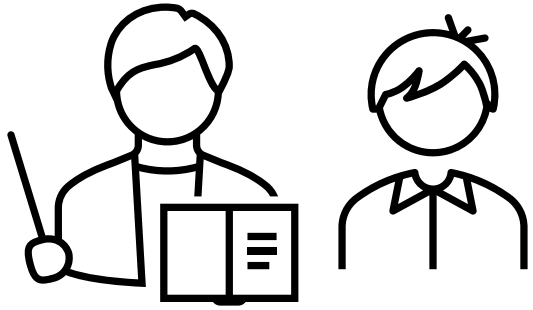


MEETING 1

ChatGPT

Fireflies



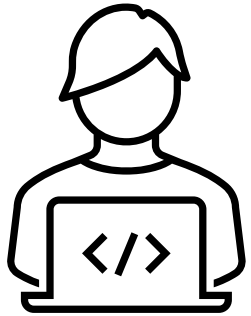


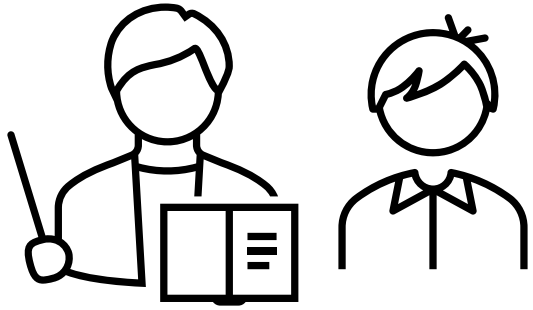
MEETING 1

ChatGPT

Fireflies

miro





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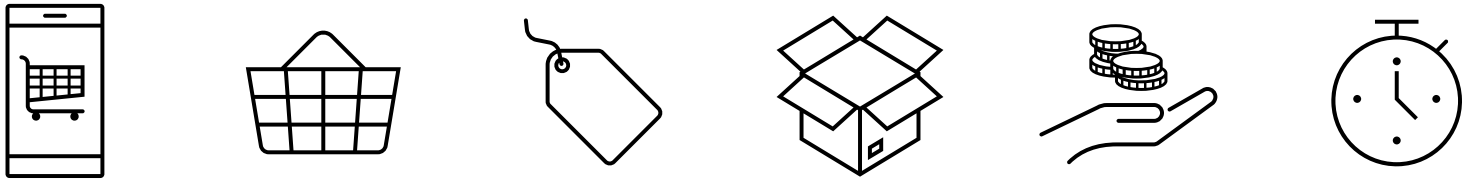

ChatGPT

MEETING 1

Fireflies

miro

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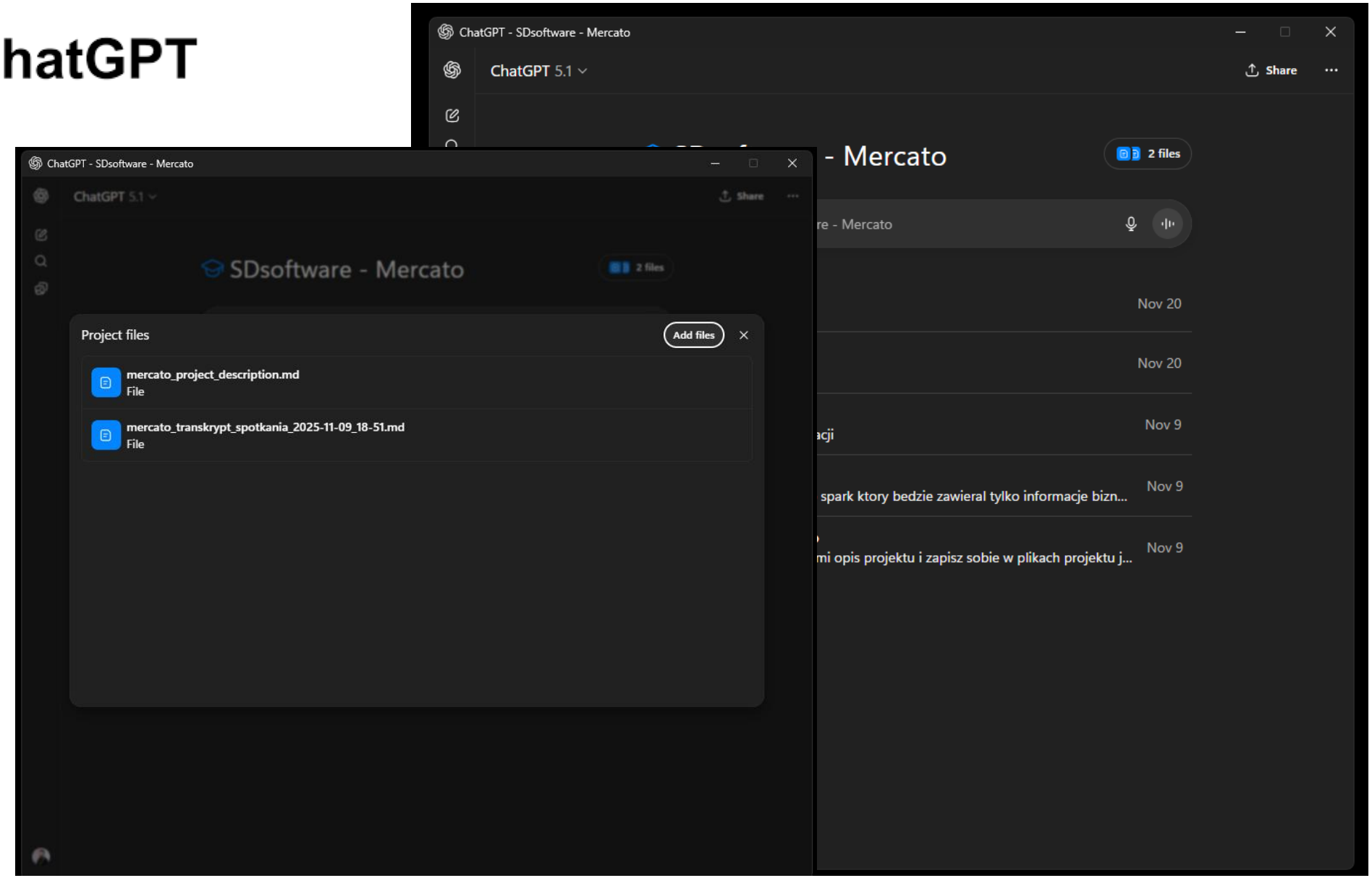
ChatGPT - SDsoftware - Mercato
ChatGPT 5.1
Share
SDsoftware - Mercato
2 files
Project files
Add files
mercato_project_description.md
File
mercato_transkrypt_spotkania_2025-11-09_18-51.md
File
- Mercato
2 files
Nov 20
Nov 20
Nov 9
spark ktory bedzie zawieral tylko informacje bizn...
Nov 9
mi opis projektu i zapisz sobie w plikach projektu j...
Nov 9

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plikacji Mercato
oduły do tej aplikacji — Nov 9

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prompt do github spark ktory bedzie zawieral tylko informacje bizn... — Nov 9

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SDsoftware - Mercato - Prompt do GitHub Spark

ChatGPT

Share

Prompt: Mercato – Marketplace for Online Stores (Business Description)

The project *Mercato* is a web-based multi-vendor marketplace connecting independent online stores in a unified system. The platform consists of three main areas:

1. **Seller Panel** – for merchants to register, add and manage products, process orders, handle payments and shipping, and view sales statistics.
2. **Buyer Portal** – for customers to browse and search products, add items to a cart, place and pay for orders, track delivery, and leave reviews.
3. **Administration Panel** – for the platform operator to manage users, listings, fees, reports, and system configuration.

The business goal is to enable small and medium-sized online stores to reach a broader audience without building their own infrastructure. Mercato's revenue model is primarily based on **sales commissions**, with optional **subscription plans** and **featured listings** in later phases.

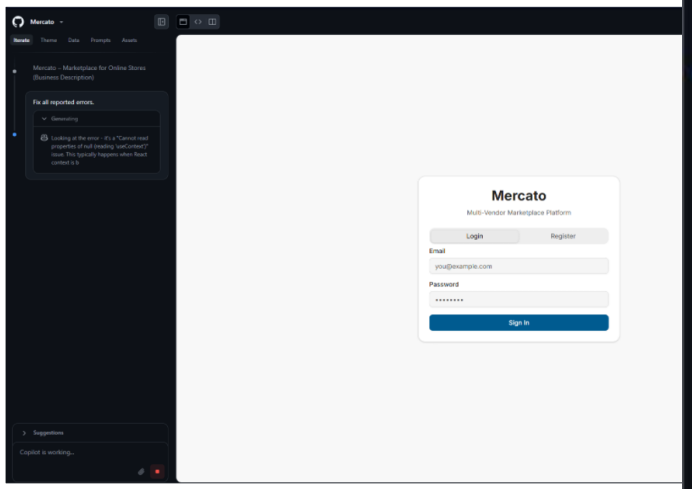
The **MVP scope** includes:

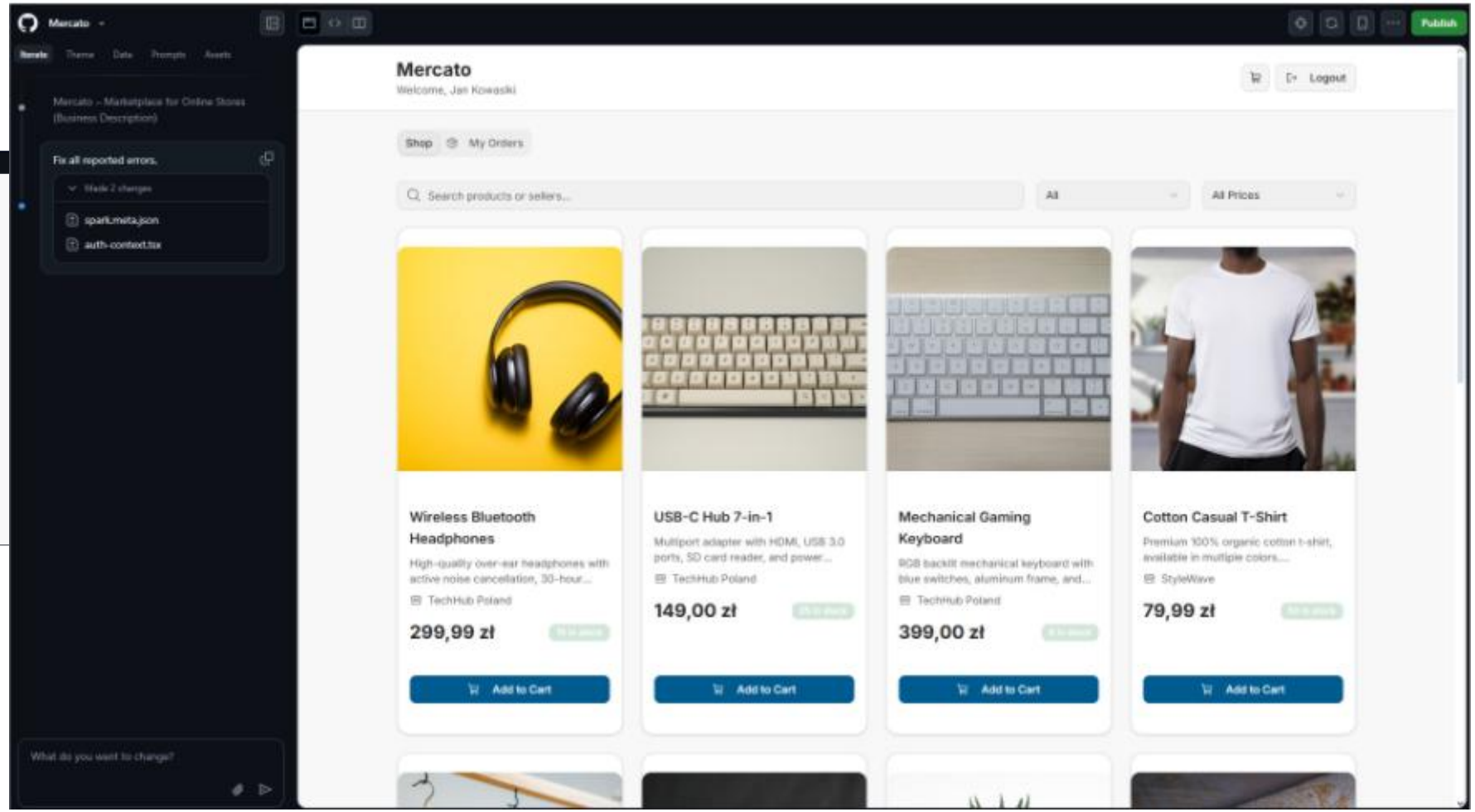
- User registration and login (buyers and sellers).
- Store profiles and product management (add, edit, delete products with descriptions, prices, and stock).
- Product browsing with categories, filters, and search.
- Shopping cart, checkout, and secure online payments via an integrated payment gateway.
- Order management (view, status update, basic returns).
- Basic notifications (email confirmations, order status).
- Simple sales dashboard for sellers and administrators.

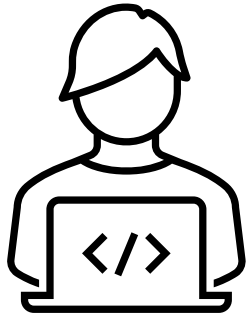
Post-MVP extensions include: reviews and ratings, loyalty programs, advanced analytics,

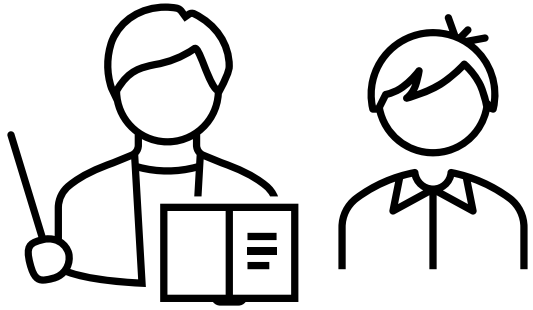
Ask anything

GitHub Spark









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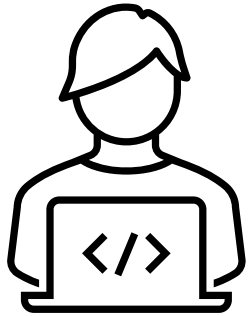

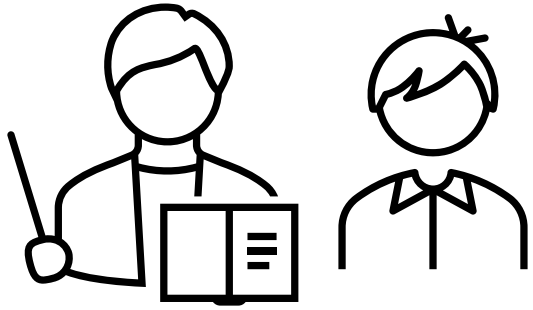

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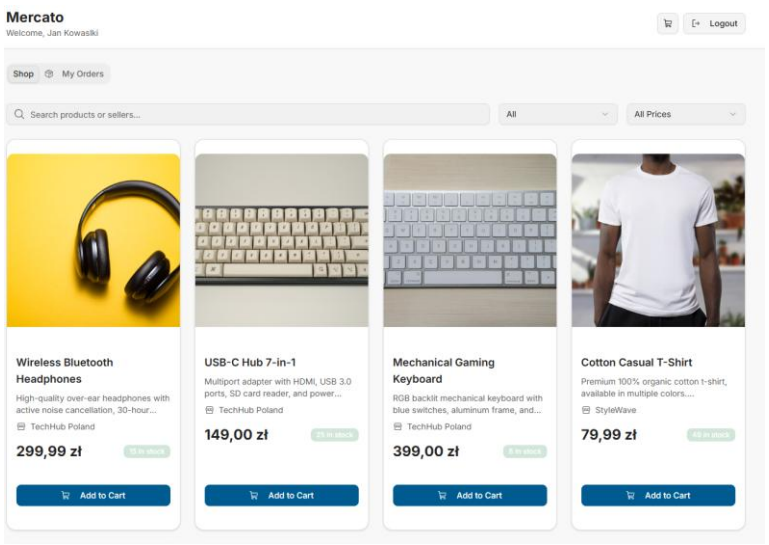
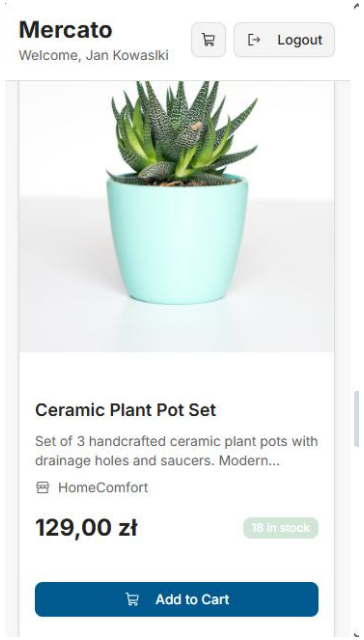
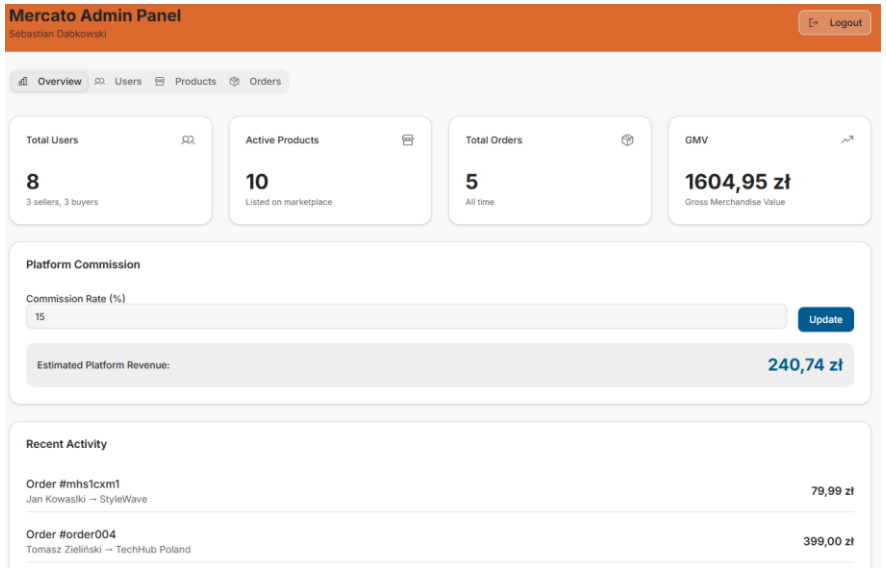

Mercato
Welcome, Jan Kowalski
Logout
Shop
My Orders
Search products or sellers...
All
All Prices
Wireless Bluetooth Headphones
High-quality over-ear headphones with active noise cancellation, 30-hour...
TechHub Poland
299,99 zł
Add to Cart
USB-C Hub 7-in-1
Multiport adapter with HDMI, USB 3.0 ports, SD card reader, and power...
TechHub Poland
149,00 zł
Add to Cart
Mechanical Gaming Keyboard
RGB backlit mechanical keyboard with blue switches, aluminum frame, and...
TechHub Poland
399,00 zł
Add to Cart
Cotton Casual T-Shirt
Premium 100% organic cotton t-shirt, available in multiple colors....
StyleWave
79,99 zł
Add to Cart


Mercato
Welcome, Jan Kowalski
Logout
Ceramic Plant Pot Set
Set of 3 handcrafted ceramic plant pots with drainage holes and saucers. Modern...
HomeComfort
129,00 zł
16 in stock
Add to Cart


Mercato Admin Panel
Sebastian Dabkowski
Logout
Overview
Users
Products
Orders
Total Users
8
3 sellers, 3 buyers
Active Products
10
Listed on marketplace
Total Orders
5
All time
GMV
1604,95 zł
Gross Merchandise Value
Platform Commission
Commission Rate (%)
15
Update
Estimated Platform Revenue:
240,74 zł
Recent Activity
Order #mhs1cxm1
Jan Kowalski → StyleWave
79,99 zł
Order #order004
Tomasz Zieliński → TechHub Poland
399,00 zł

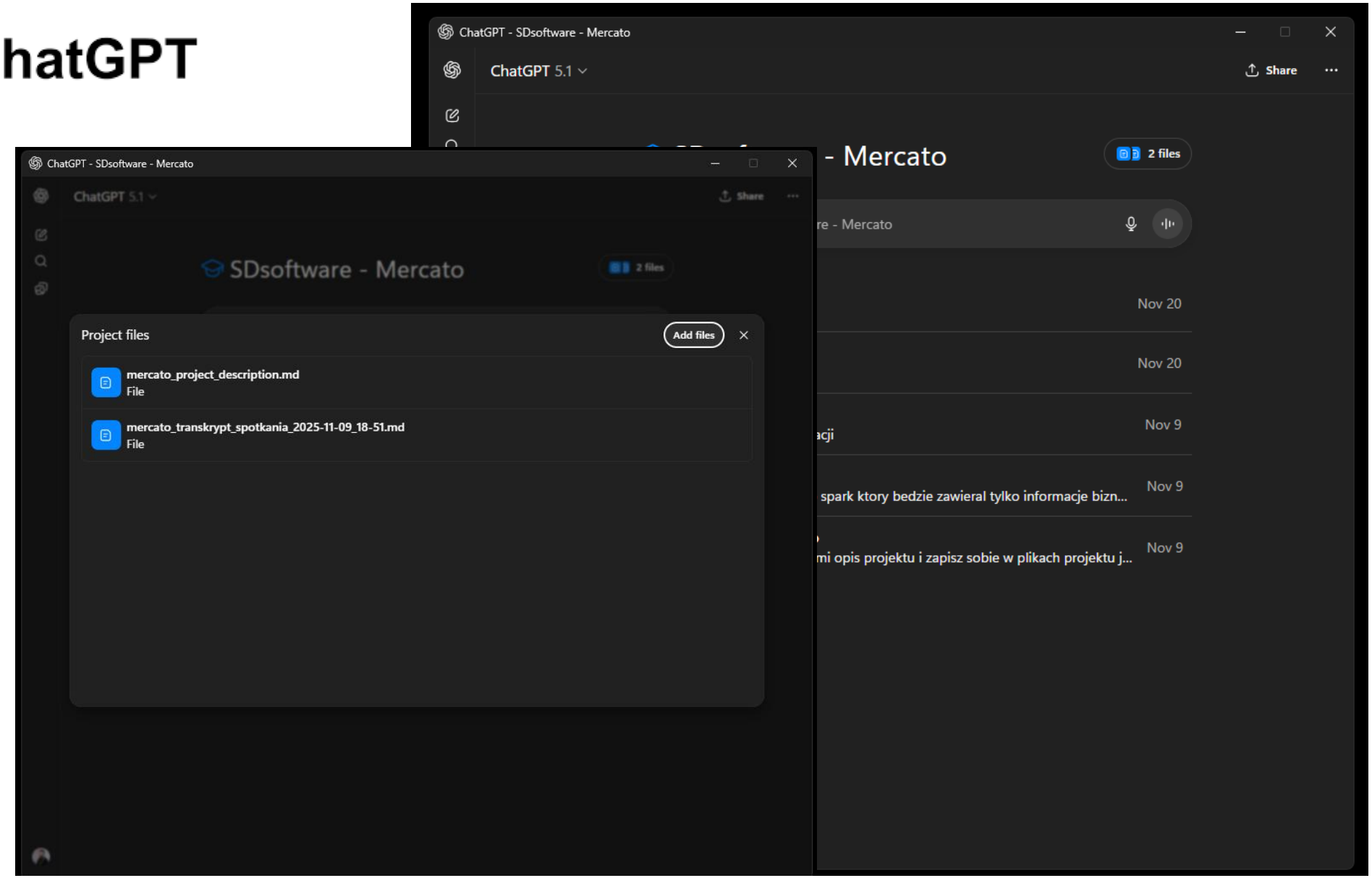
ChatGPT

MEETING 2

Fireflies

miro

ChatGPT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2 files
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mercato_project_description.md
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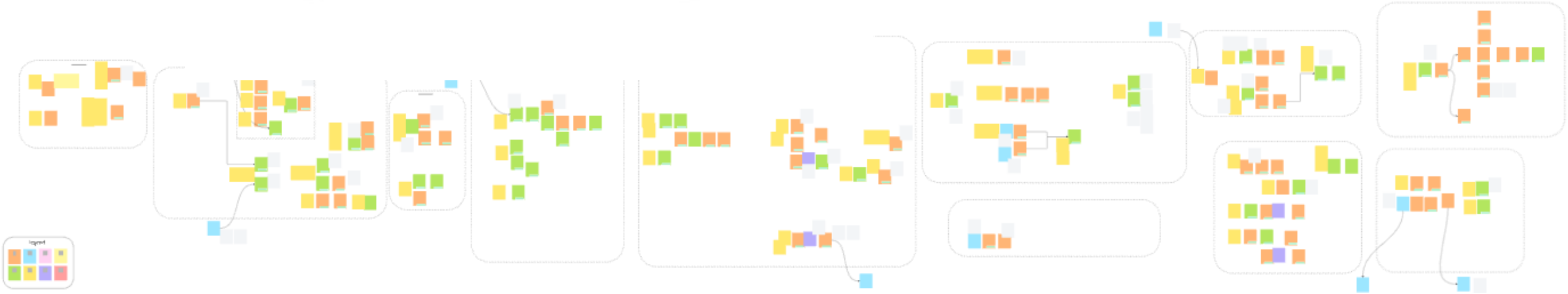
mercato_transkrypt_spotkania_2025-11-09_18-51.md
File
- Mercato
2 files
Nov 20
Nov 20
Nov 9
spark ktory bedzie zawieral tylko informacje bizn...
Nov 9
mi opis projektu i zapisz sobie w plikach projektu j...
Nov 9

EVENT STORMING

EVENT STORMING



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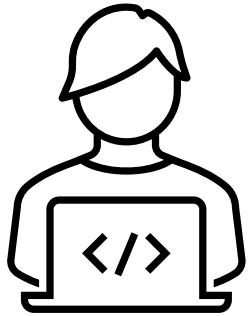

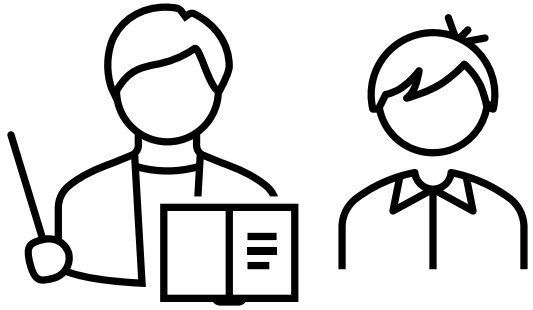

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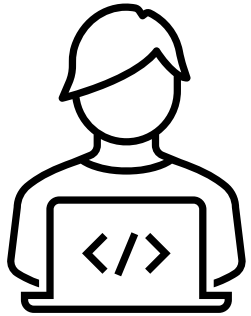
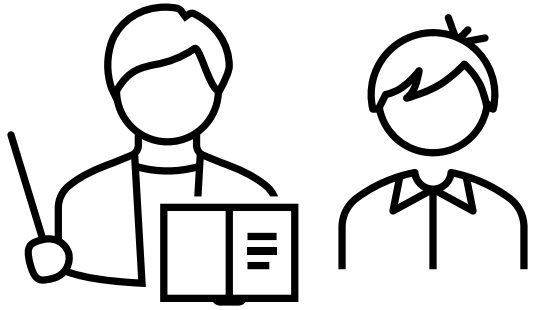
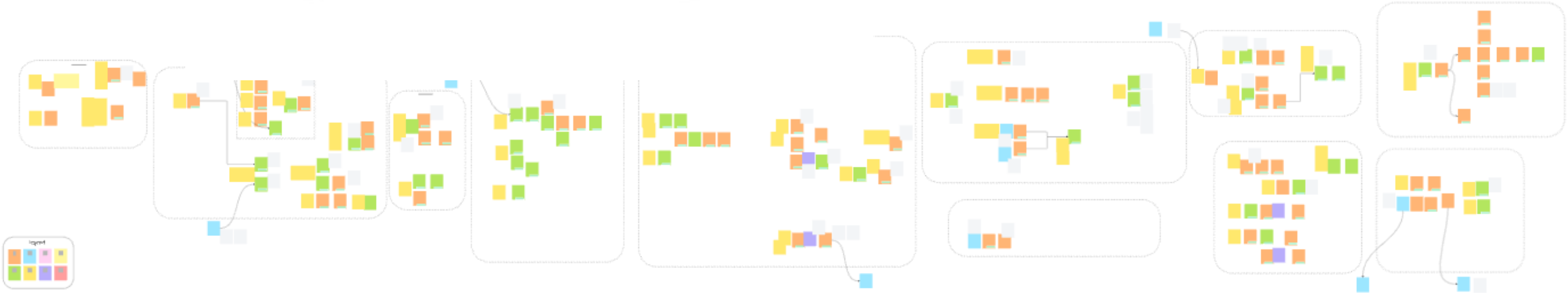


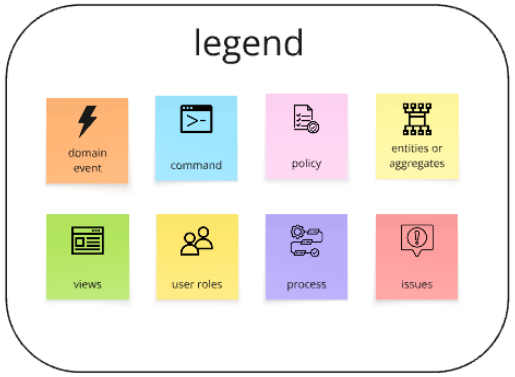


MEETING 3

EVENT STORMING

EVENT STORMING

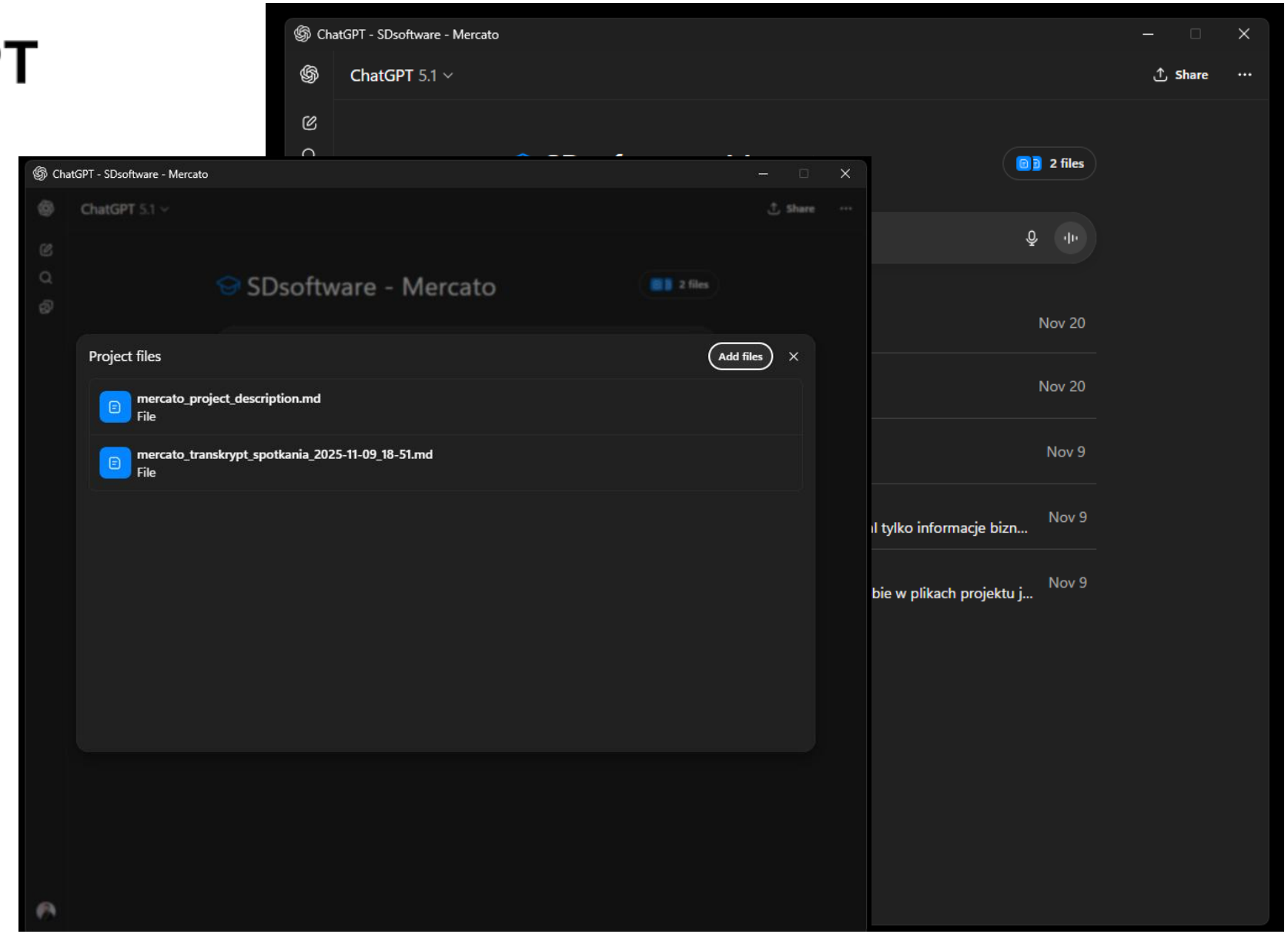




miro

Fireflies

ChatGPT


ChatGPT - SDsoftware - Mercato
ChatGPT 5.1
Share
2 files
Nov 20
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Nov 9
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Nov 9
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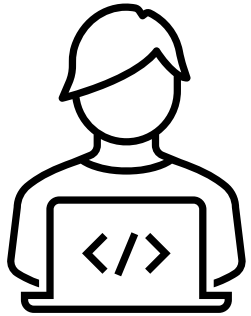
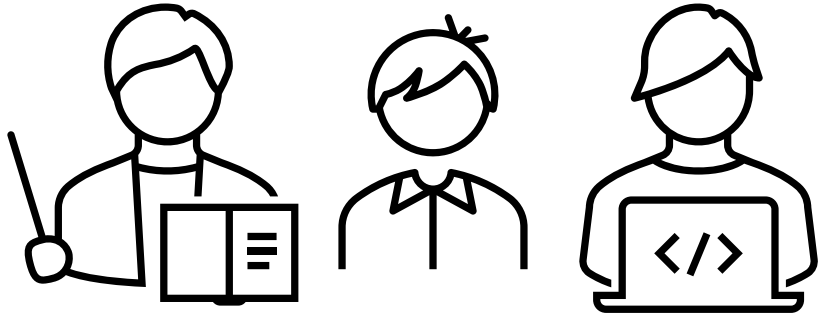
Nov 9
ChatGPT - SDsoftware - Mercato
ChatGPT 5.1
Share
SDsoftware - Mercato
2 files
Project files
Add files
mercato_project_description.md
File
mercato_transkrypt_spotkania_2025-11-09_18-51.md
File

Architecture Characteristics

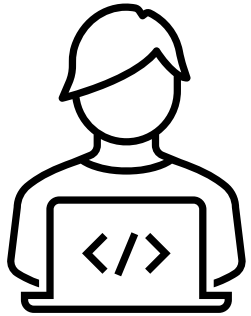
Simplicyty ★★★★★ Scalibility ★★★★★ Maintainability ★★★★★ evolvability ★★★★★

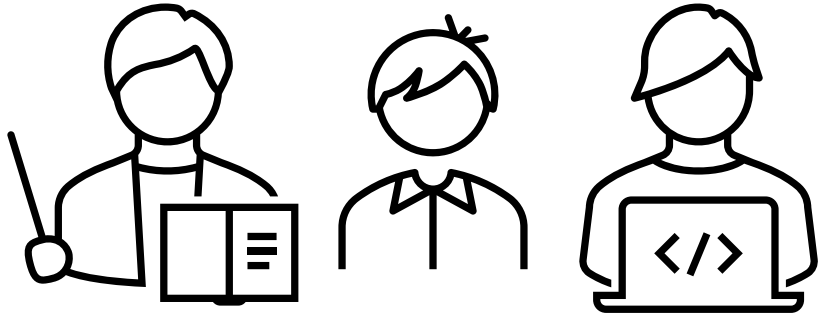
MEETING 4





MEETING 4





MEETING 4

Architecture Characteristics

Simplicyty ★★★★★ Scalibility ★★★★★ Maintainability ★★★★★ evolvability ★★★★★

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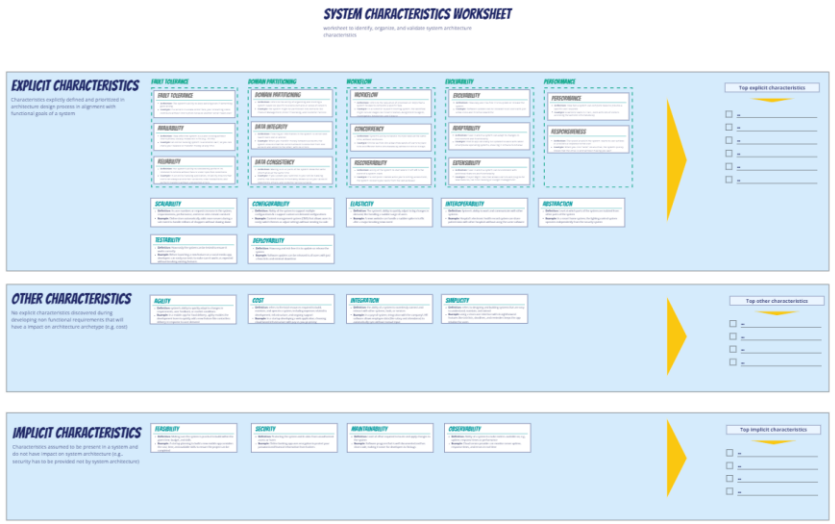
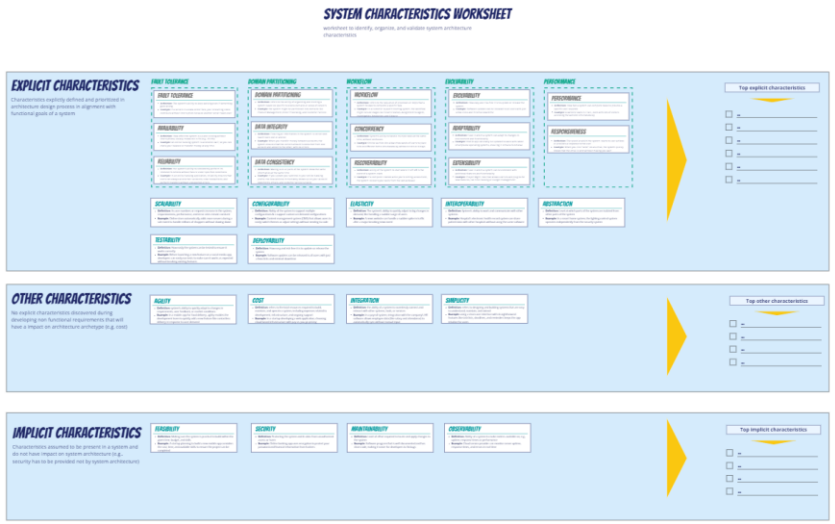


Selected Architecture(s):

	☐	3 ☐	4 ☐	9 ☒	6 ☐	☐	☐	☐
	layered	modular monolith	microkernel	microservices	service-based	service-oriented	event-driven	space-based
partitioning	technical	domain	domain	domain	domain	technical	technical	technical
cost	$	$	$	$$$$$	$$	$$$$	$$$	$$$$
maintainability	★	★★	★★★	★★★★★	★★★★	★	★★★	★★★
testability	★★	★★	★★★	★★★★★	★★★★	★	★★	★
deployability	★	★★	★★★	★★★★★	★★★★	★	★★★	★★★
simplicity	★★★★★	★★★★★	★★★★	★	★★★	★	★★	★
scalability	★	★	★	★★★★★	★★★	★★★	★★★★	★★★★★
elasticity	★	★	★	★★★★	★★	★★★	★★★	★★★★★
responsiveness	★★★	★★★	★★★	★★	★★★	★★	★★★★★	★★★★★
fault-tolerance	★	★	★	★★★★★	★★★★	★★★	★★★★★	★★★
evolvability	★	★	★★★	★★★★★	★★★★	★	★★★★★	★★★
abstraction	★	★	★★★	★	★	★★★★★	★★★★	★
interoperability	★	★	★★★	★★★	★★	★★★★★	★★★	★★

Architecture Characteristics

Simplicyty ★★★★★ Scalibility ★★★★★ Maintainability ★★★★★ Evolvability ★★★★★



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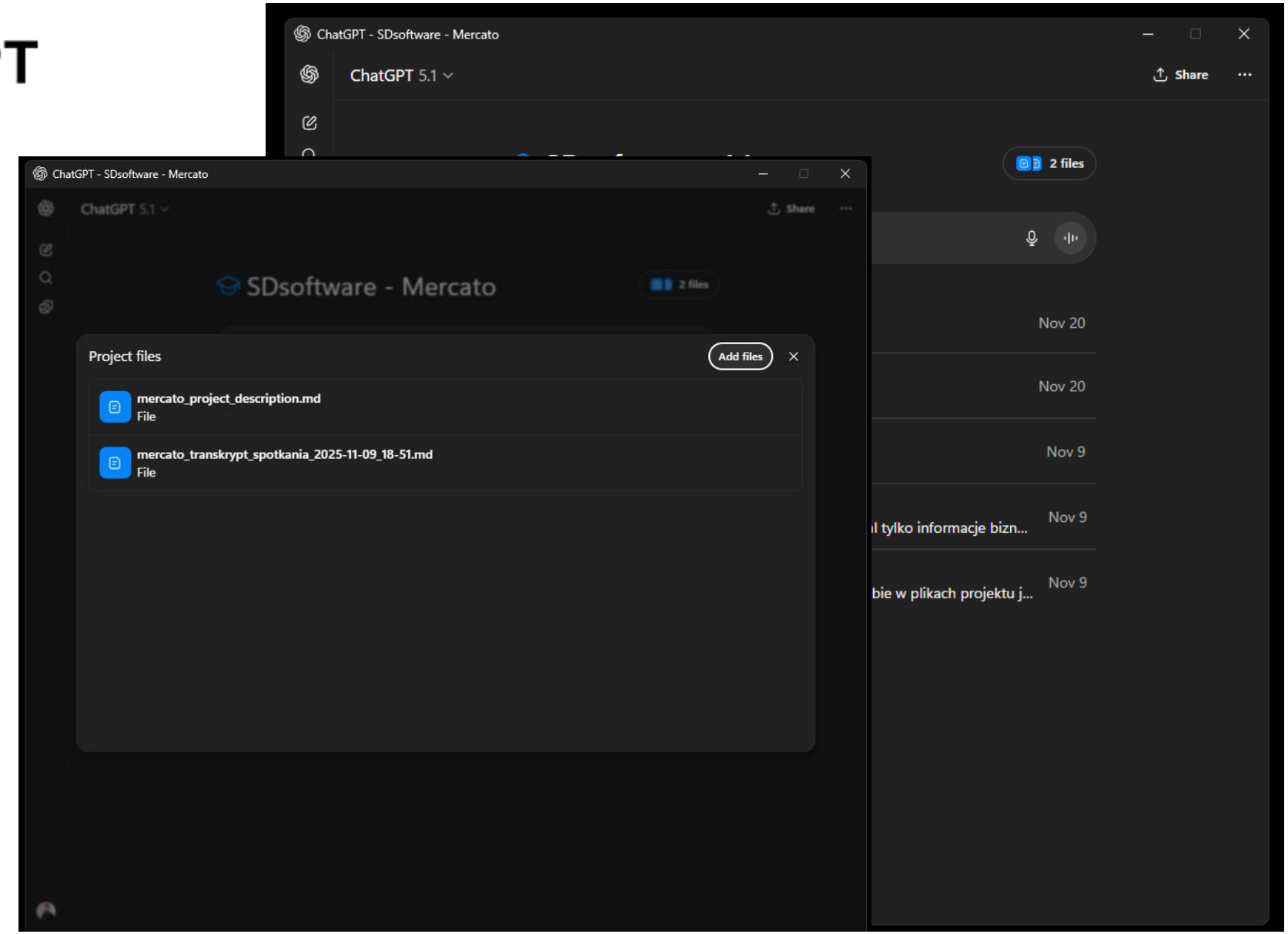
deployability	★	★★	★★★	★★★★★	★★★★	★	★★★	★★★
simplicity	★★★★★	★★★★★	★★★★	★	★★★	★	★★	★
scalability	★	★	★	★★★★★	★★★	★★★	★★★★	★★★★★
elasticity	★	★	★	★★★★	★★	★★★	★★★	★★★★★
responsiveness	★★★	★★★	★★★	★★	★★★	★★	★★★★★	★★★★★
fault-tolerance	★	★	★	★★★★★	★★★★	★★★	★★★★★	★★★
evolvability	★	★	★★★	★★★★★	★★★★	★		
abstraction	★	★	★★★	★	★	★★★★★		
interoperability	★	★	★★★	★★★	★★	★★★★★		

miro

Fireflies

ChatGPT


ChatGPT - SDsoftware - Mercato
ChatGPT 5.1
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2 files
Nov 20
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ChatGPT 5.1
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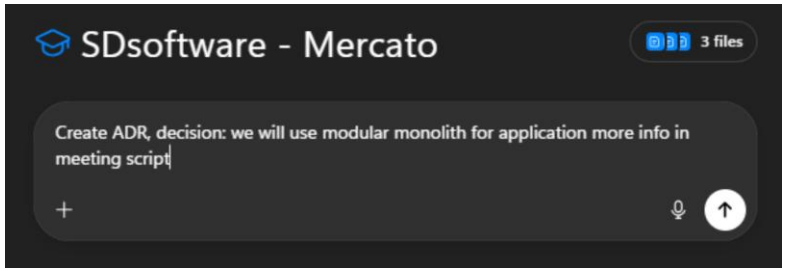
2 files
Project files
Add files
mercato_project_description.md
File
mercato_transkrypt_spotkania_2025-11-09_18-51.md
File

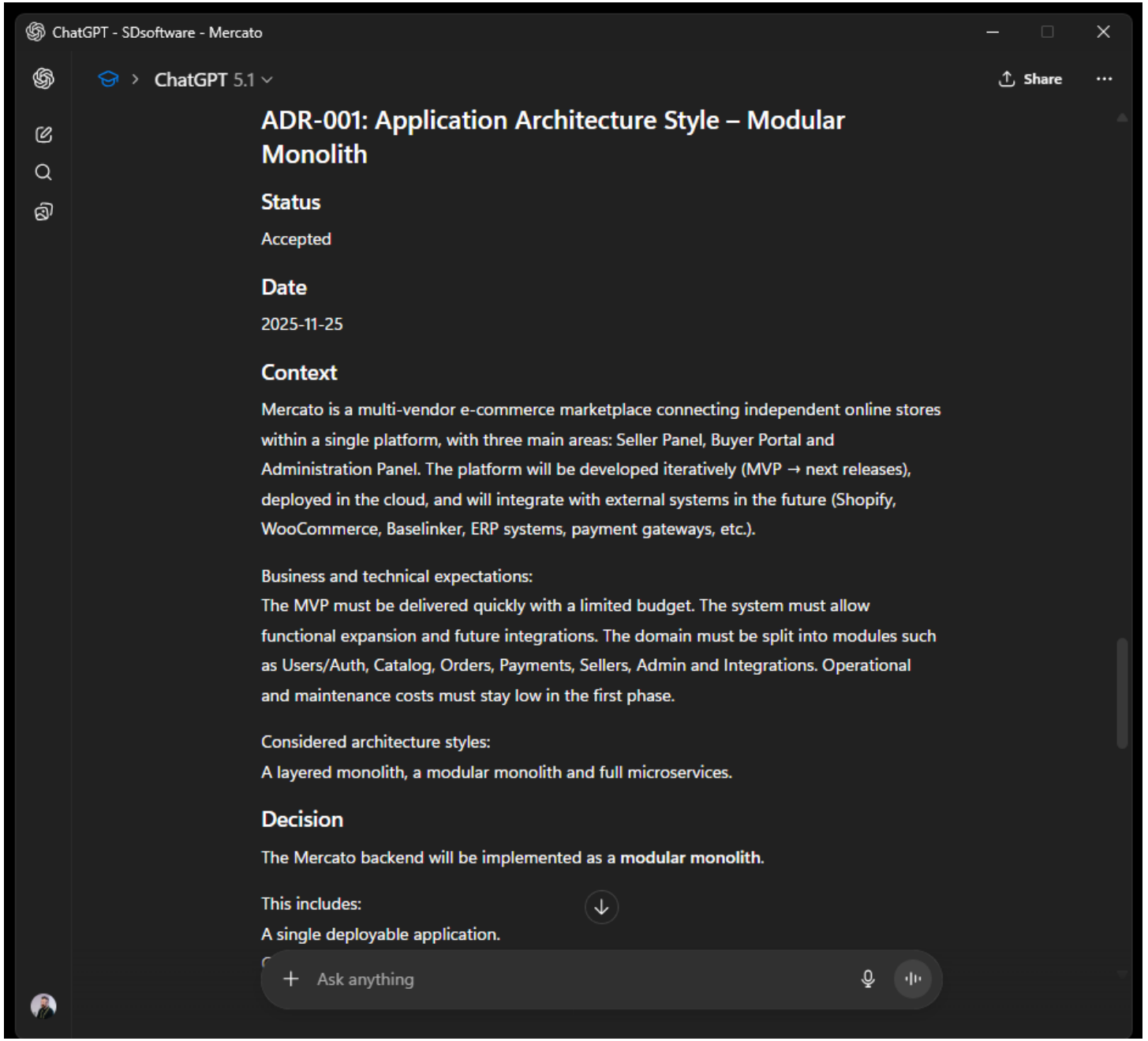
ADR

Architecture decision record

ADR

Architecture Decision Record





ARCHITECTURE

Business Scope

Features | Revenue | Costs | How? | What? | Who?

EVENT STORMING

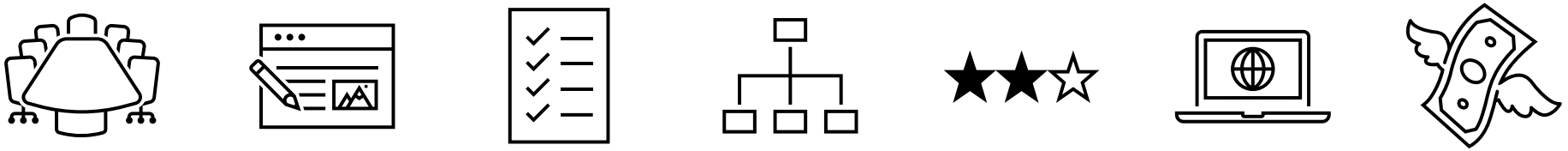
ARCHITECTURE CHARACTERISTICS

SIMPLICYTY ★★★★★ SCALIBILITY ★★★★★ MAINTAINABILITY ★★★★★ EVOLVABILITY ★★★★★

ADR

ARCHITECTURE DECISION RECORD

ARCHITECTURE



CODING

SCOPE

ARCHITECTURE

TEAM

SCOPE

PRD.MD

README.MD

TASKS

ChatGPT

Scope

ChatGPT - SDsoftware - Mercato

ChatGPT 5.1

Share

spotkania.

Epic 1: Identity & Access Management

Features:

1. Buyer registration and login with email and password.
2. Seller registration and login with email and password.
3. Social login for buyers via Google and Facebook, Apple later. mercato_transkrypt_spotkania_20...
4. Account verification flow, including email confirmation and optional KYC for sellers.
5. Password reset and change password flow.
6. Session management and logout.
7. Role model: buyer, seller, marketplace admin, finance/admin back office.
8. Basic account security settings (2FA ready, login history view in later phase).

Epic 2: Seller Management

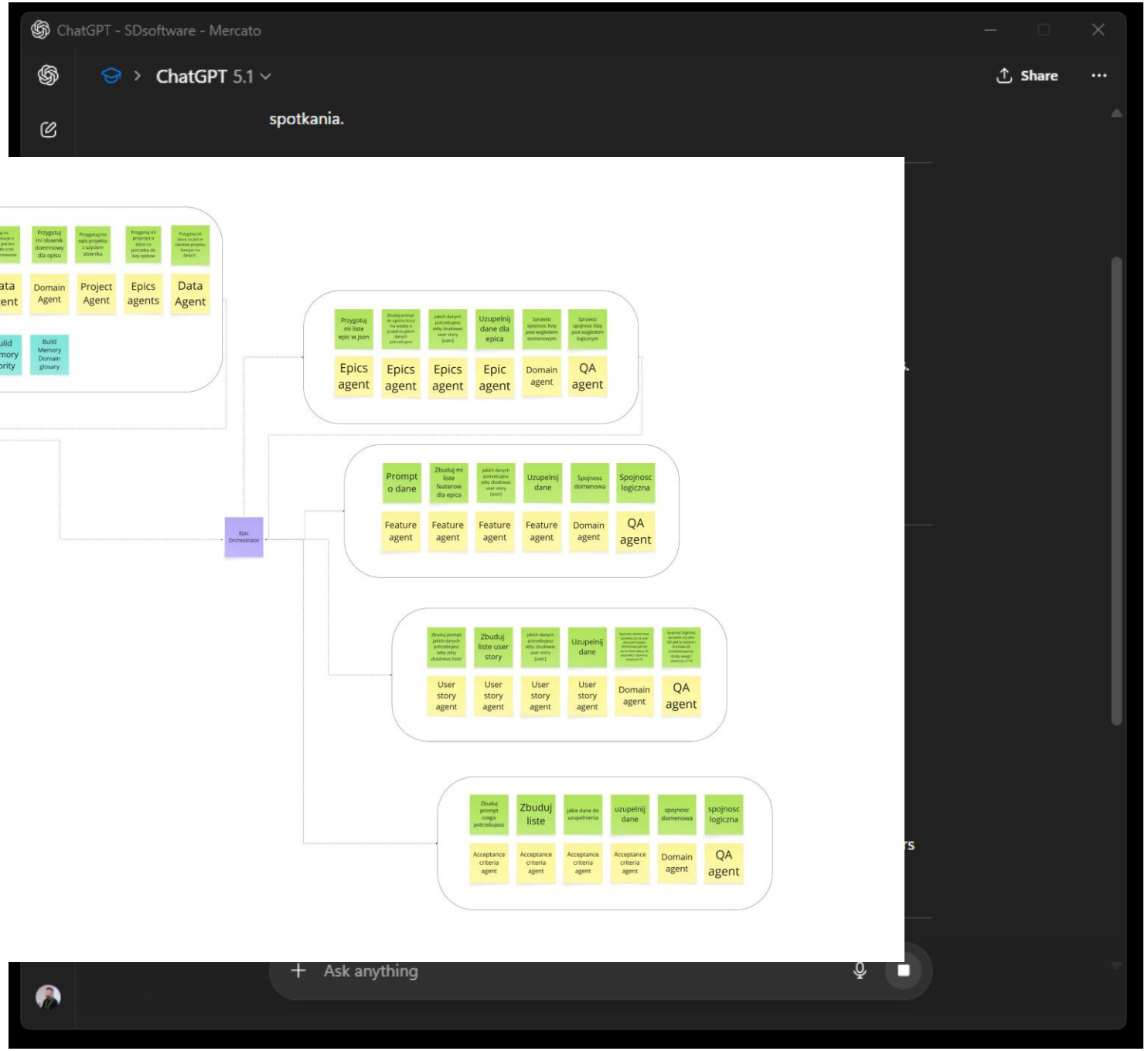
Features:

1. Seller onboarding wizard for new stores.
2. Seller profile page with store name, logo, description and contact details.
3. Store verification process with company data, NIP or personal ID for individuals. mercato_transkrypt_spotkania_20...
4. Store profile public view under Mercato domain path (mercato.pl/store-name). mercato_transkrypt_spotkania_20...
5. Seller account settings: bank account, payout preferences, invoicing details.
6. Seller user management in later phase: owner and additional users with roles (offers manager, orders operator, accountant). mercato_transkrypt_spotkania_20...

Ask anything

ChatGPT

Scope



ChatGPT

Scope

ChatGPT - SDsoftware - Mercato

ChatGPT 5.1

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Ask anything

ARCHITECTURE

ARCHITECTURE.MD

TEAM

INSTRUCTION.MD

AGENT.MD

PRD.MD

TASKS

README.MD

ARCHITECTURE.MD

INSTRUCTION.MD

AGENT.MD

AI Agent compare

Experiment 2

1. Prepare user story
2. Prepare architecture
    - N-Layer
3. Implement
    - CODEX
    - CLAUDE
    - COPILOT

ASP.NET™

AI Agent compare

CODEX CLAUDE COPILOT

AI Agent compare

CODEX

mercatoApp11 Public
coding by: Codex
C# Updated 2 weeks ago

CLAUDE

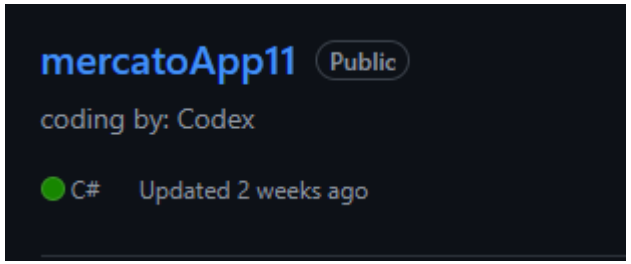
mercatoApp12 Public
coding by: Cloude
C# MIT License Updated last week

COPILOT

MercatoApp13 Public
coding by: Copilot
C# Updated 5 days ago

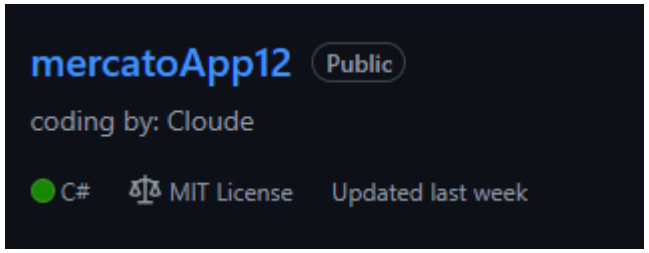
AI Agent compare

CODEX



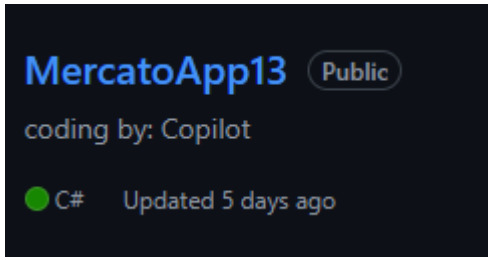
GPT-5.1-Codex-Max

CLAUDE



Claude Haiku 4.5

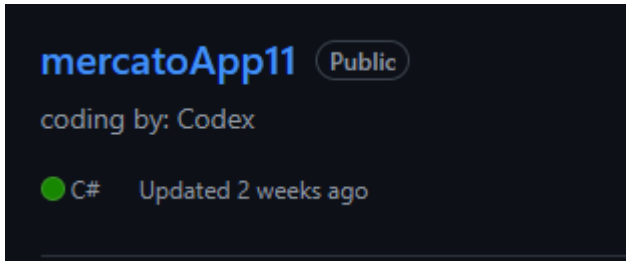
COPILOT



Coding Agent model

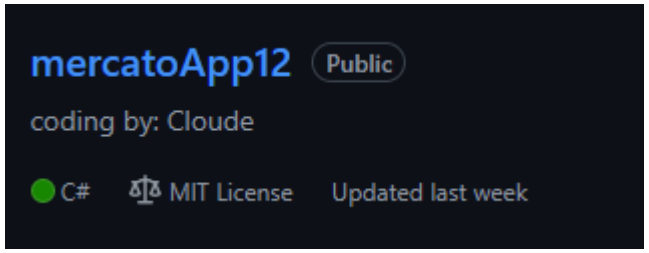
AI Agent compare

CODEX



GPT-5.1-
Codex-Max

CLAUDE



Claude
Haiku 4.5

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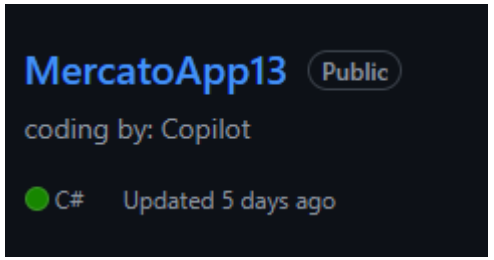

Claude
Sonnet 4.6

Cost

	CODEX	CLAUDE	COPILOT
Effort	22h 44min	30h 21min	19h 02min
Total cost	$ 15,62	$ 13,82	$ 11,03
PR	133	133	133
Cost PR	$ 0,12	$ 0,10	$ 0,08
Cost h	$ 0,69	$ 0,46	$ 0,58

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Effort	22h 44min	30h 21min	19h 02min
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Cost PR	$ 0,12	$ 0,10	$ 0,08
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Agent compare

		CODEX	CLAUDE	COPILOT
Price				
	Cost PR	$ 0,12	$ 0,10	$ 0,08
	Cost h	$ 0,69	$ 0,46	$ 0,58
Quality (Sonar)				
	Software	11,87	8,95	10,83
	Code attribute	10,77	8,06	9,11
	Type	10,77	7,98	9,11
Quality (Snyk)				
	All	1,96	19,04	5,54

Agent compare

		CODEX	CLAUDE	COPILOT
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	Cost PR	$ 0,12	$ 0,10	$ 0,08
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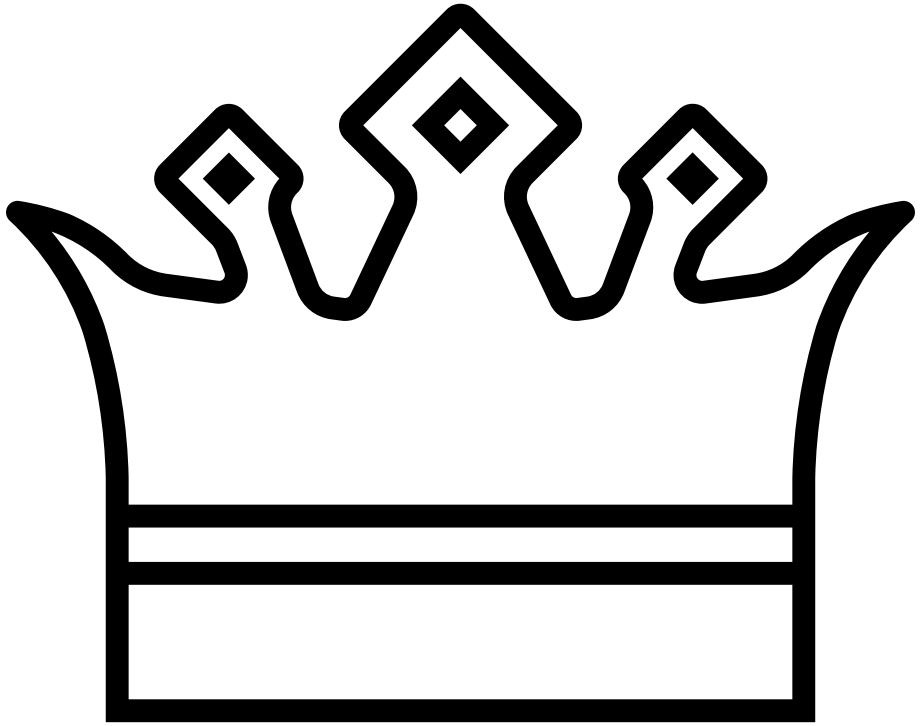

CLAUDE CODE

LEARNINGS

LEARNINGS

Tech Stack – big impact

LEARNINGS

Tech Stack – big impact

Merging – conflicts

LEARNINGS

Tech Stack – big impact

Merging – conflicts

Missing URLs in system

LEARNINGS

Tech Stack – big impact

Merging – conflicts

Missing URLs in system

Database migration – Fail

Architecture

Business Scope

Features | Revenue | Costs | How? | What? | Who?

EVENT STORMING

ARCHITECTURE CHARACTERISTICS

SIMPLICYTY ★★★★★ SCALIBILITY ★★★★★ MAINTAINABILITY ★★★★★ EVOLVABILITY ★★★★★

ADR

ARCHITECTURE DECISION RECORD

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ADR

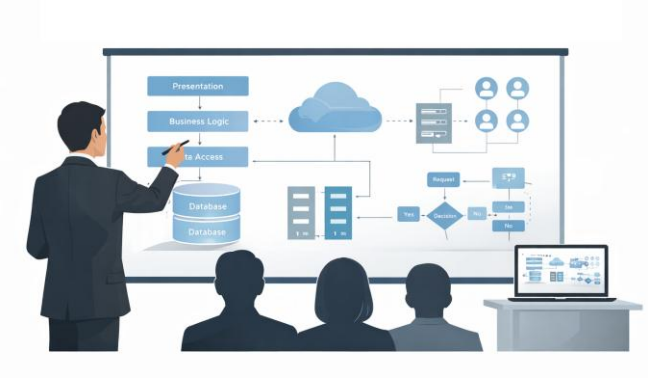
ARCHITECTURE DECISION RECORD

The **C4 model**

for visualising software architecture

TEAM TOPOLOGIES

FITNESS FUNCTIONS



Architectural Kata

AI made coding cheap.

Architecture, process
and business partnership
create real value.

THANK YOU

Sebastian Dąbkowski

THANK YOU

Q & A

Sebastian Dąbkowski